Special Notice

Request for Information:
Embedding Electricity Generation from Heat in Microsystems
DARPA-SN-25-70
April 30, 2025

Defense Advanced Research Projects Agency
Microsystems Technology Office
675 North Randolph Street
Arlington, VA 22203-2114

Request for Information (RFI)
Special Notice DARPA-SN-25-70

Embedding Electricity Generation from Heat in Microsystems
Defense Advanced Research Projects Agency (DARPA)
Microsystems Technology Office (MTO)

Posting Date: April 30, 2025
Responses Due: May 26, 2025, at 5:00 p.m. Eastern Time (ET)
Technical POC: Dr. Yogendra Joshi, MTO Program Manager
E-mail: DARPA-SN-25-70@darpa.mil

RFI DESCRIPTION:
The Defense Advanced Research Projects Agency (DARPA) Microsystems Technology Office (MTO) seeks information on suitable and/or commercial microsystem(s) for possible demonstration of dissipated heat to electricity conversion at the chip/substrate/package levels. DARPA is interested in all types of microsystems for possible use as an exemplar to demonstrate technologies for embedded dissipated heat to electricity conversion.

The goals of this Request for Information (RFI) include, but are not limited to:

- Identification of suitable and/or commercially available microsystem(s) including, but not limited to, graphics processing unit, central processing unit, neural processing unit, monolithic microwave integrated circuit, and laser diode relevant for potential performers to utilize as an exemplar.
- Identification of all relevant intellectual property (IP) associated with the microsystem, such as physical mapping of back-end of the chip/package/interposer/substrate, keep-out areas for electrical routing, power map of front-end transistors/devices for designating hotspot regions, etc. relevant for potential performers to strategize implementation of technologies within the package.
- Identification of organizations interested in providing the relevant information identified above to prospective performers under a related potential DARPA program (assume the potential for non-disclosure agreements (NDAs) if necessary).
- Identification of organizations interested in providing on the order of one to three of the identified chip/microsystem packages to prospective performers under a related potential DARPA program.
- Identification of potential applications and impacts of using electricity converted from dissipated heat in chip/microsystem packages.

BACKGROUND:
Commercial and defense microsystems consume useful electrical power and, consequently, generate waste heat. At present, this waste heat is transported through varying pathways to a heat sink and rejected into the atmosphere. If this heat were to be corralled and transported from its source to an embedded reclamation device, the recovered electricity has the potential to transform commercial microsystem technology and power infrastructure. Refer to DARPA-SN-25-69 on sam.gov for information on electrical power generation from microsystem waste heat through corralling dissipated heat and reclaiming it as useful electricity. This RFI is not seeking solutions for the challenges discussed in DARPA-SN-25-69.

REQUESTED INFORMATION:
DARPA seeks information from microsystems manufacturers that may be relevant to a potential DARPA program on reclamation of waste heat into electricity. Responses are welcome from all capable sources including, but not limited to, private or public companies, individuals, universities, university-affiliated research centers, not-for-profit research institutions, Federally Funded Research and Development Centers (FFRDCs), and U.S. Government-sponsored labs. Responses to the RFI should address the following items of interest, pertaining to the goals identified above in the RFI Description.

1. Describe the information you would be willing to provide to prospective performers for use under a potential DARPA program related to the Background section above.
2. Identify the target chip/substrate/microsystem package you would be willing to provide to prospective performers under a potential DARPA program related to the Background section above and address the following:
   a. Relevant technical IP information listed under “RFI Description”
   b. Multiple counts (up to three) of the chip/substrate/microsystem package
   c. Required Non-Disclosure Agreement (NDA) considerations
3. Describe some possible applications and impacts of using electricity converted from dissipated heat in the chip/microsystem packages you identified in response to 2. above.

SUBMISSION INSTRUCTIONS:
Responses to this RFI should be submitted no later than 5:00 p.m. ET on May 26, 2025.

Unclassified responses to this RFI should be submitted to DARPA-SN-25-70@darpa.mil. NO CLASSIFIED INFORMATION SHOULD BE SENT TO DARPA-SN-25-70@darpa.mil.

To the maximum extent possible, respondents should submit non-proprietary information. If proprietary information is submitted, it must be appropriately and specifically marked. It is the respondent’s responsibility to clearly define to the Government what is considered proprietary data. Any proprietary information should be clearly labeled as “Proprietary.” DARPA will disclose submission contents only for the purpose of review by DARPA staff, other Government agencies, or DARPA Support Contractors/SETAs.

NOTE: DARPA intends to conduct individual discussions with respondents as necessary to gain a full understanding of the technical and partnership models submitted. DARPA will contact respondents individually via e-mail.

FORMAT INSTRUCTIONS:
Responses to the RFI should be concise. Respondents should submit a single integrated response addressing the requested information above. DARPA will only review responses submitted as an

unprotected Microsoft Word/PowerPoint document or PDF file. Each response is limited to no more than 6 pages using 11-point font and 1-inch margins on 8.5-inch by 11-inch page size. Any submitted material in excess of these limits may not be reviewed.

Responses should adhere to the following formatting instructions:

1. Cover page (1 page)
    a. Organization(s)
    b. Respondent's technical and administrative points of contact (names, addresses, phone and fax numbers, and email addresses)
2. An executive summary (1 page maximum)
3. Responses to requested information described above (1 page maximum)
4. Specification sheet(s) for target chip(s)/microsystem package(s) (1 page each, 3 pages maximum)

DARPACONNECT:
Entities who have not worked with DARPA before are encouraged to learn more about DARPAConnect, an initiative established to facilitate collaboration between DARPA and potential performers. The DARPAConnect team offers customized support, resources, and guidance on how to prepare your ideas for high-impact conversations with DARPA program managers. Please visit DARPAConnect.us to access a digital hub of online resources, including a curriculum for self-paced learning, personalized support, and in-person and virtual events. In addition to the self-paced online materials, the DARPAConnect team is able to schedule one-on-one conversations to discuss your specific ideas, questions, and paths to DARPA. You can use the contact form at DARPAConnect.us or email the DARPAConnect team directly at darpaconnect@darpa.mil to request assistance.

ADMINISTRATIVE:
This announcement contains all information required to submit a response. No additional forms, kits, or other materials are needed. All administrative and technical questions should be directed to DARPA-SN-25-70@darpa.mil. Please refer to the Special Notice number (DARPA-SN-25-70) in all correspondence.

This RFI is issued solely for information and program planning purposes and does not constitute a formal solicitation for proposals or proposal abstracts; any sent will be disregarded. In accordance with FAR 15.201(e), responses to this notice are not offers and cannot be accepted by the Government to form a binding contract. Submission of a response is strictly voluntary and is not required to propose to subsequent Announcements (if any) or Solicitations (if any) on this topic. DARPA will not provide reimbursement for costs incurred in responding to this RFI. Respondents are advised that DARPA is under no obligation to acknowledge receipt of the information received or provide feedback to respondents with respect to any information submitted under this RFI.